

Information Retrieval Notes

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Chapter 1

Introduction and Architecture of Information Retrieval

1.1 Definition and Motivations

Information Retrieval (IR) is the scientific and engineering discipline concerned with representing, storing, organizing, and accessing large collections of predominantly unstructured or semi-structured data to satisfy an **information need** expressed by a user [1, 2].

Definition 1.1.1 (Information Retrieval (Manning et al.)). Information Retrieval is finding material of an unstructured nature (usually text) that satisfies an information need from within large collections stored on computers.

IR is not limited to general-purpose Web search engines (Google, Bing): it underpins e-commerce and marketplace search, enterprise/desktop search, **RAG** (*Retrieval-Augmented Generation*) systems feeding Large Language Models with contextually relevant external documents, and recommender engines.

1.1.1 From Web Search to Agentic Search

The rise of conversational LLMs has not made Information Retrieval obsolete but has broadened its impact: modern LLMs continuously rely on core IR competencies (semantic query understanding, matching and ranking, contextual retrieval) to overcome their *knowledge cutoff*. In the **RAG** paradigm, the system intercepts the user’s request, queries an IR engine, injects the retrieved documents into the LLM’s context, and generates a response supported by verifiable citations.

Definition 1.1.2 (Agentic Search). **Agentic Search** is a dynamic retrieval process in which an intelligent agent autonomously decides which subsequent searches to conduct, based on partial evidence gathered: it inspects sources, identifies knowledge gaps (*information gaps*), formulates refined follow-up queries, and iterates until sufficient evidence has converged.

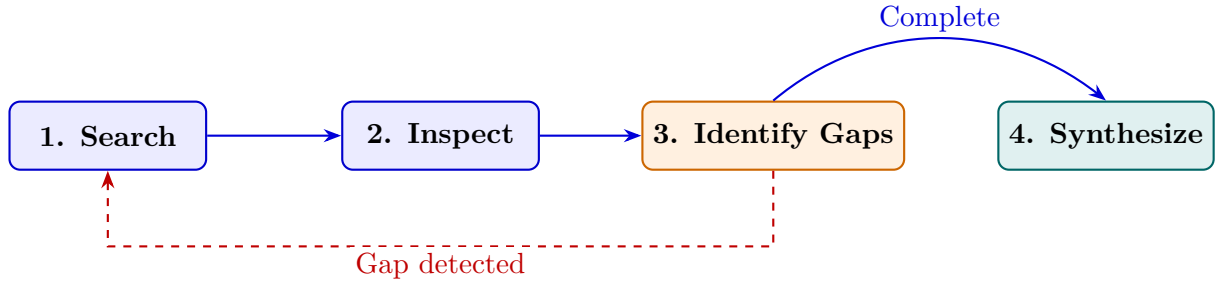


Figure 1.1: Adaptive feedback loop characterizing Agentic Search.

1.1.2 Information Retrieval vs. Data Retrieval (Databases)

Dimension	Data Retrieval (SQL Databases)	Information Retrieval
Data structure	Rigidly schematized (tables, types, constraints)	Unstructured or semi-structured (free text, HTML)
Query language	Formal and deterministic (SQL)	Natural or keyword-based, often ambiguous
Matching	Exact Boolean	Partial and probabilistic (degrees of relevance)
Ordering	None unless explicit (ORDER BY)	Fundamental, driven by relevance estimation
Error tolerance	None	High (typos, synonymy, polysemy)

Table 1.1: Comparison between traditional Data Retrieval and Information Retrieval.

Relevance: The Cornerstone of IR

While in a database a record either satisfies or violates a Boolean predicate, in Information Retrieval the core challenge is estimating the degree of **relevance** with respect to the user's information need: subjective, context-dependent, and evolving over time.

1.2 The Two Macro-Objectives: Effectiveness and Efficiency

A large-scale retrieval system pursues two objectives simultaneously, often in tension:

1. **Effectiveness:** how accurately the ranking satisfies the user's information need.
2. **Efficiency:** computational cost, throughput, and response latency, typically on the order of 10–50 milliseconds per query over collections of billions of documents.

The efficiency of a modern search engine depends on its interaction with the underlying hardware architecture: pipelining and superscalar processors, memory hierarchy and access locality, SIMD vectorization, and parallel/distributed computing across clusters (*sharding*).

1.3 Architecture of an Information Retrieval System

An IR system enforces a strict decoupling between the **Offline Pipeline** (indexing and ranking model training) and the **Online Pipeline** (low-latency query serving), as depicted in Figure 1.2.

1.3.1 Text Processing Pipeline

Before indexing, every text stream goes through: **tokenization** (segmentation into lexical tokens), **normalization/case folding** (lowercasing, diacritics), **stop word filtering** (removal of high-frequency, low-discriminative words), and **stemming/lemmatization** (reducing an inflected form to its root or canonical lemma, e.g. "searching" → "search").

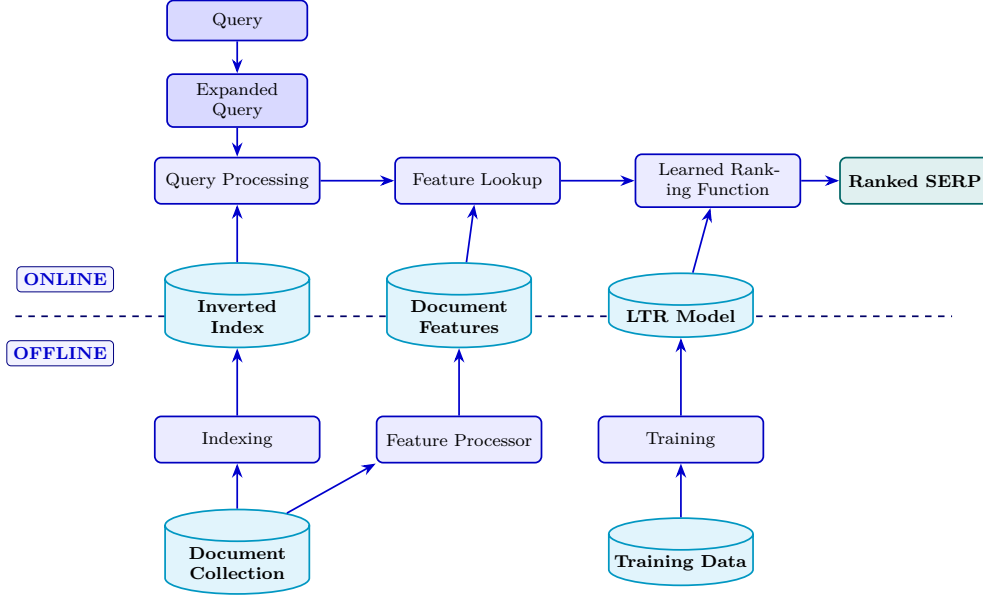


Figure 1.2: Grid architecture of a modern search engine (Online vs. Offline pipeline).

1.4 Language Properties and Retrieval Models

- **Zipf's Law:** the frequency $f(r)$ of a term is inversely proportional to its rank r in the frequency table, $f(r) \propto 1/r$: a few very frequent words account for most occurrences, while most terms occur only rarely (*long tail*).
- **Heaps' Law:** vocabulary size grows as a power law of the total number of words N in the collection, $|V| = k \cdot N^\beta$ (with $0.4 \leq \beta \leq 0.6$): the vocabulary never fully saturates.

The main retrieval models are the **Boolean Model** (exact match via logical operators, no relevance score), the **Vector Space Model** (queries and documents as TF-IDF weighted vectors, relevance via cosine similarity), and the **Probabilistic BM25 Model**, the de facto standard for lexical retrieval:

$$\text{score}_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{tf(t, d) \cdot (k_1 + 1)}{tf(t, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (1.1)$$

where $\text{IDF}(t) = \ln\left(\frac{N-df_t+0.5}{df_t+0.5} + 1\right)$ weighs term specificity, k_1 (typically 1.2–2.0) controls term-frequency saturation, and $b \in [0, 1]$ (typically ≈ 0.75) normalizes document length $|d|$ against the average length avgdl .

1.5 The Inverted Index

The **Inverted Index** reverses the logical orientation of document storage, realizing a mapping $t \mapsto \mathcal{P}_t$, where $\mathcal{P}_t$ is the sorted list of documents containing term t [1].

Definition 1.5.1 (Inverted Index). An Inverted Index consists of a **dictionary** (the sorted set of unique terms, with document frequency df_t and a pointer to the posting list) and, for each term, a **posting list**: an array sorted by increasing DocID, whose entries may include metadata such as local term frequency (tf) and positional offsets.

Example 1.5.1 (Reference Toy Collection). Consider a small collection of three documents about indexing and text search. Figure 1.3 shows the Inverted Index built for a few key terms: **a** appears in documents 1, 2, 3; **an** only in document 1; **index** and **inverted** in documents 1, 2; **query** only in document 3.

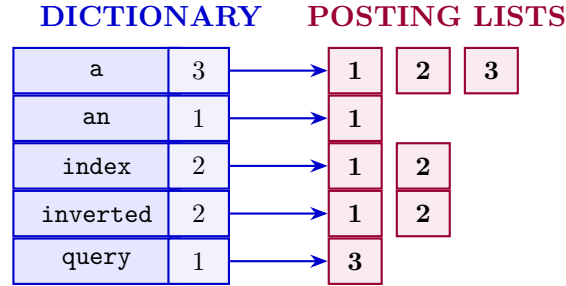


Figure 1.3: Structure of the Inverted Index: dictionary and posting lists.

Boolean constructs (AND, OR, NOT) and phrase search map directly to the advanced search interface of commercial search engines (Table 1.2).

IR Boolean Operator	Google Advanced Search Field	Syntax
Conjunction (AND)	<i>all these words</i>	<code>tri-colour rat terrier</code>
Phrase search	<i>this exact word or phrase</i>	<code>"rat terrier"</code>
Disjunction (OR)	<i>any of these words</i>	<code>miniature OR standard</code>
Negation (NOT)	<i>none of these words</i>	<code>-rodent -"Jack Russell"</code>
Numeric range	<i>numbers ranging from...to</i>	<code>10..35 kg</code>

Table 1.2: Mapping between IR Boolean operators and Google Advanced Search.

1.6 Index Compression Techniques

Inverted Index compression reduces memory footprint (RAM/disk), converting an I/O bottleneck into a CPU decompression load, which is generally much faster.

1.6.1 Binary Interpolative Coding

Binary Interpolative Coding (Moffat and Stuiver [3]) compresses strictly increasing sequences of integers (the DocIDs of a posting list) by recursively partitioning them: given a sublist of n integers known to lie within $[l, h]$, the middle element x_m (with $m = \lfloor (n + 1)/2 \rfloor$) is encoded using $\lceil \log_2 R \rceil$ bits, where $R = h - l - n + 1$ is the number of admissible values, and the recursion proceeds separately on the two halves. When DocIDs are densely clustered, R tends toward 1 and many elements are encoded at near-zero cost.

1.6.2 Dictionary-Based Compression: LZ77

For general text, the **Lempel-Ziv '77** (LZ77) algorithm compresses via a sliding window, replacing lexical repetitions with triples (offset, length, next_char) referring to earlier occurrences in the window. This family of algorithms underlies widely adopted tools such as `gzip`, `7-Zip`, and `LZ4`.

1.7 Learning-to-Rank (LTR) and the Non-Differentiability Challenge

Candidates retrieved by the Inverted Index are passed to a machine learning re-ranking stage known as **Learning-to-Rank** (LTR) [4]. The training dataset associates each query with a list of documents carrying graded relevance judgments $y \in \{0, \dots, 4\}$ (from irrelevant to perfectly relevant); a model computes a score $s_i = f_\theta(\vec{x}_{q,d_i})$ for each query-document pair, and the final list is obtained by sorting in decreasing order of score.

The Cornerstone of LTR

The goal is to learn the ranking, not the label: the objective is not to accurately estimate the single value y , but the optimal relative ordering, maximizing cumulative metrics such as **NDCG** or **MAP**.

Because the sorting operation is discrete and step-wise, the gradient of ranking metrics with respect to model parameters is zero almost everywhere: gradient descent cannot be applied directly. The literature has developed three families of solutions:

1. **Pointwise:** treats each document as an independent instance (regression or classification on y), ignoring the inherently comparative nature of ranking.
2. **Pairwise** (e.g. RankNet): compares pairs of documents, modeling the probability that d_i precedes d_j with a differentiable sigmoid.
3. **Listwise** (e.g. LambdaMART): optimizes the entire list via surrogate gradients λ_{ij} , weighted by the actual NDCG change resulting from swapping d_i and d_j ; combined with gradient-boosted decision trees, it was for over a decade the standard in commercial search engines.

1.8 The Shift Toward Neural Information Retrieval

Since 2018, Transformer-based models (BERT) have ushered in the era of Neural Information Retrieval [5, 6], pre-trained via **Masked Language Modeling**. Neural retrieval approaches fall into two families:

- **Interaction-based (Cross-Encoders):** query and document are concatenated and jointly processed by a Transformer, capturing deep

interactions at high computational cost ($\mathcal{O}(|\mathcal{D}| \times \text{Transformer})$); used only as second-stage re-rankers over a small candidate set.

- **Representation-based (Bi-Encoders):** query and document are independently projected into a vector space by two distinct encoders; document vectors can be pre-computed offline, reducing online matching to a dot product or an approximate nearest-neighbor search.

Among representation-based models, three families of embeddings are distinguished: **dense single-vector** (a single dense vector per document/query, queried with k-ANN graph indexes such as HNSW), **dense multi-vector** (e.g. ColBERT [5], one vector per token with *Late Interaction*/MaxSim aggregation, $S(q, d) = \sum_{t_q \in q} \max_{t_d \in d} (\vec{E}_{t_q} \cdot \vec{E}_{t_d}^T)$), and **sparse single-vector** (e.g. SPLADE [6], interpretable by design and more space/time efficient than multi-vector).

k-Approximate Nearest Neighbor Search

For collections of millions of documents represented as high-dimensional dense vectors, exact retrieval of the k most similar to the query is computationally prohibitive: index structures for **approximate nearest-neighbor search** (k -ANN) are used instead, trading a minimal loss of accuracy for orders-of-magnitude speedups. **TusKANNy** (<https://tuskanny.com/>), developed by ISTI-CNR and the University of Pisa, is a high-performance open-source example written in Rust.

Chapter 2

Evaluation of Information Retrieval Systems

2.1 Search Engine Architecture

Before diving into evaluation methodology, it is instructive to understand the high-level block diagram of a modern search engine, since each architectural component introduces measurable quality dimensions.

The main components are:

- **Web Crawler / Document Collection:** An automated agent that continuously discovers and downloads documents from the Web (or a closed corpus), feeding the indexing pipeline.
- **Parsing & Tokenization:** Documents are linguistically analyzed: HTML is stripped, text is tokenized into terms, stop-words may be removed, and stems or lemmas are computed.
- **Inverted Index:** The core data structure mapping each term to the list of documents containing it (posting list), together with positional and frequency information.
- **Query Processing & Query Expansion:** An incoming user query is tokenized with the same pipeline, possibly enriched with synonyms, related terms, or session context to improve recall.
- **Feature Extraction:** For each candidate document, a rich feature vector is assembled (BM25 score, PageRank, freshness, anchor text signals, click history, ...).

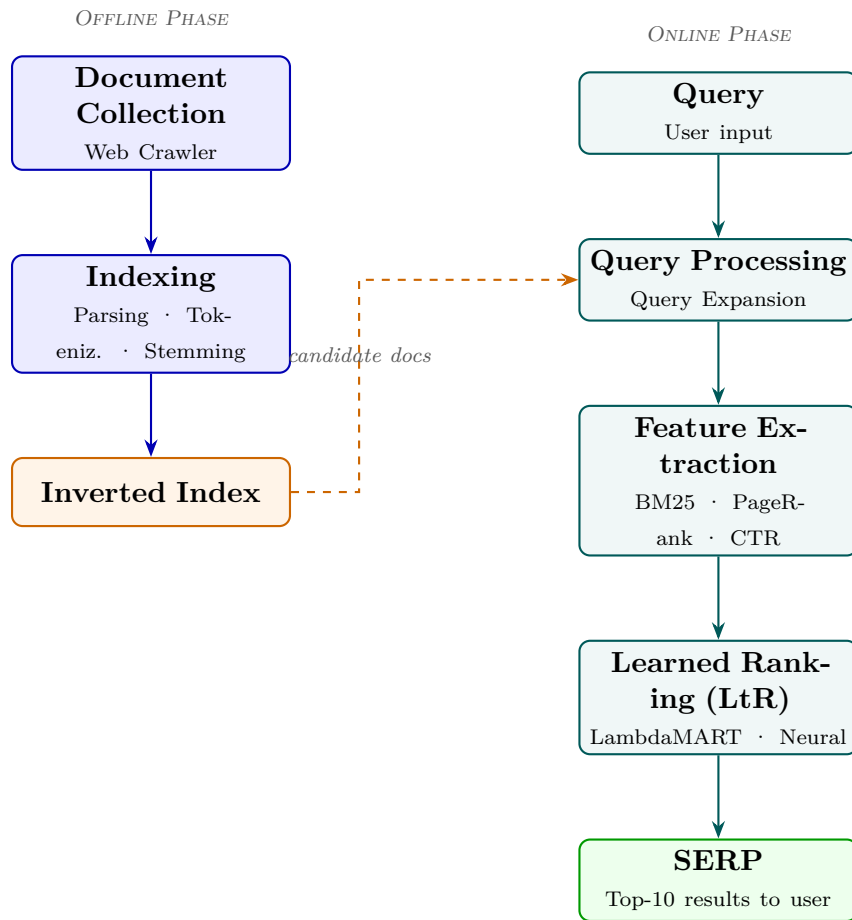


Figure 2.1: High-level block diagram of a modern search engine: offline indexing pipeline (left) and online query processing and ranking pipeline (right).

- **Learned Ranking Function (Learning to Rank):** A machine-learned model (e.g., LambdaMART, neural ranker) combines the features to produce a final relevance score used to sort the SERP.
- **SERP (Search Engine Results Page):** The final ordered list of results displayed to the user, possibly augmented with featured snippets, knowledge panels, and ads.

2.2 Evaluating Search Engine Quality

How do we evaluate such a complex, multi-component system? The answer lies in separating two complementary dimensions—**efficiency** and **effectiveness**—each requiring distinct measurement methodologies [1, 2].

- **Efficiency** concerns the *speed and resource cost* of processing: how fast can we index documents, and how quickly can we answer queries at scale?
- **Effectiveness** concerns the *quality of the answers*: are the returned documents genuinely useful to the user?

Rigorous performance evaluation is essential for designing, fine-tuning, and deploying Information Retrieval systems [1, 2]. Search engine quality is thus evaluated across these two complementary dimensions:

1. Efficiency Metrics:

- *Indexing throughput:* Number of documents processed and incorporated per hour.
- *Incremental indexing:* Ingesting new document streams (e.g., +10 000 products daily) without downtime.
- *Search latency and throughput:* Query processing response time (demanding 95th/99th percentile latencies within 10–50 ms) and CPU/memory utilization under workloads of thousands of queries per second over multi-million or multi-billion document collections.

2. Effectiveness Metrics and Auxiliary Services:

- Quality of related product recommendations, query spell-checking accuracy, and snippet generation readability.
- **Intrinsic ranking effectiveness:** Measures the engine’s core capability to satisfy users by ranking relevant documents at the top of the Search Engine Results Page (SERP).

2.3 Measuring User Happiness

In commercial Web search engines, the overarching objective is maximizing **User Happiness**. Because happiness is an internal cognitive state that cannot be directly queried, production environments rely on observable online behavioral proxies:

- **Clicks and Click-Through Rate (CTR):** The volume and proportion of clicks received by returned results.
 - *Caveat:* Misleading titles and clickbait summaries artificially inflate clicks without truly satisfying the user.
 - *Zero-Click Searches:* When is “no clicks” good news? For factual queries (e.g., weather forecasts, flight schedules, currency conversions, or dictionary definitions), answers rendered directly via *Featured Snippets* or *Knowledge Panels* satisfy the user immediately without requiring any downstream click.
- **Purchases and Economic Conversions:** In e-commerce, the percentage of search sessions resulting in transactions and total monetary spend serve as key indicators.
- **Retention and Repeat Visits:** The fraction of users returning to the search engine within a week or month.
- **Dwell Time:** The duration an attendee spends exploring a clicked page before navigating back to the SERP (*bouncing back*). Extended dwell time indicates genuine engagement, whereas near-instant returns signal dissatisfaction.

Evaluation in the Era of Generative AI and LLMs

The rise of generative conversational engines (such as ChatGPT, SearchGPT, and Perplexity) alters traditional CTR and Dwell Time dynamics: users consume direct synthesized summaries rather than navigating outgoing links. Consequently, factual grounding, hallucination minimization, and citation fidelity supplant traditional link click telemetry.

Several important debates have emerged in the research community regarding how to evaluate these new systems:

- **Number of conversational turns as a failure proxy (raised by student Dino):** In a generative dialogue system, a high number of follow-up turns might indicate that the initial response was unsatisfactory—the user is forced to re-ask or clarify. Conversely, resolving the need in a single turn could signal high effectiveness. This is an emerging behavioral proxy with no settled consensus.
- **Prompting variance issue (raised by student Gabriele):** Unlike a keyword query submitted to a classical IR engine, the phrasing of a prompt to a generative LLM can dramatically change the output quality. This makes reproducible evaluation extremely difficult: the same underlying information need can be expressed in thousands of ways, each yielding a different response.
- **Ontological difference from classic IR (raised by student Bruno):** Traditional IR retrieves *existing* documents; generative LLMs *synthesize* new text. This is a fundamentally different epistemic act—the system is no longer a retrieval engine but a generator, making relevance judgments harder to ground in objective fact.
- **Retrieval-Augmented Generation (RAG):** One prominent engineering paradigm to mitigate hallucinations is RAG, in which a generative model is provided with retrieved documents as grounding context before generating its response. RAG partially bridges the gap between classical IR and LLMs, but introduces new evaluation challenges: both the retrieval step and the generation step must be assessed independently and jointly.

2.4 The Cranfield Paradigm and Offline Test Collections

While online signals offer valuable telemetry, they are noisy, subject to position bias, and unsuitable for comparing algorithmic improvements offline prior to production deployment.

To situate the Cranfield approach historically: Cyril Cleverdon conducted his seminal experiments in the **1960s**, working with *physical* library systems—card catalogues and aeronautical engineering records at the Cranfield Institute of Technology—long before digital text retrieval was computationally feasible. His insight was to construct a controlled experimental collection with exhaustive relevance judgments, providing the first rigorous laboratory for IR comparison. This setting, humble by today’s standards, established a scientific template still followed six decades later.

To establish reproducible, controlled evaluation, IR research adheres to the **Cranfield Paradigm**, pioneered by Cyril Cleverdon in the 1960s [7].

Definition 2.4.1 (Test Collection). A standard Information Retrieval test collection is a formal triplet $\mathcal{T} = (\mathcal{D}, \mathcal{Q}, \mathcal{J})$ consisting of:

1. A benchmark document collection $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$;
2. A benchmark suite of queries $\mathcal{Q} = \{q_1, q_2, \dots, q_M\}$;
3. A set of relevance judgments (*qrels*) $\mathcal{J} = \{(q, d) \mapsto r_{q,d}\}$, assigning a ground-truth relevance assessment (binary or graded) to each query-document pair.

2.4.1 Combinatorial Scale Bottlenecks

Exhaustively assessing all query-document pairs is physically infeasible for realistic corpus sizes. Consider a representative benchmark with:

$$|\mathcal{D}| = 5 \times 10^6 \text{ documents, } |\mathcal{Q}| = 50\,000 \text{ queries}$$

The full Cartesian product comprises:

$$|\mathcal{D}| \times |\mathcal{Q}| = 5 \times 10^6 \times 50\,000 = 2.5 \times 10^{11} \text{ pairs } (0.25 \text{ trillion judgments})$$

Assuming an experienced human assessor evaluates one document in 2.5 seconds:

$$\begin{aligned} \text{Total Time} &= 2.5 \times 10^{11} \times 2.5 \text{ s} = 6.25 \times 10^{11} \text{ seconds} \\ &\approx 173\,611\,111 \text{ person-hours} \end{aligned}$$

This would require tens of thousands of human working years! Consequently, benchmark datasets must restrict assessment to a sampled subset of candidate documents.

2.4.2 Origins of Test Queries

A natural question is: *where do the queries in a benchmark test suite come from?* The answer differs significantly depending on the evaluation context:

- **Web search engines — Query Logs:** Large-scale industrial benchmarks (such as those used in Microsoft TREC tracks or the AOL query log dataset) sample queries directly from *real user query logs*. This ensures that the test suite faithfully reflects the actual distribution of user information needs, including rare and long-tail queries.
- **Classic laboratory scenarios — Expert-drafted lists:** In pre-Web IR benchmarks (Cranfield, TREC ad hoc tracks), domain experts *manually composed* a set of information need statements (called *topics*). Each topic is a verbose narrative describing the underlying information need, which is then abbreviated into a short keyword query for system input.
- **What is *not* done — Random queries from documents:** One might naively suggest generating queries by sampling random terms from documents in the collection. This approach is **explicitly avoided** in IR evaluation: randomly sampled terms do not resemble real user intents, would introduce severe artificial biases, and would not generalize to the retrieval scenarios that actually matter to end users.

2.4.3 Crowdsourcing: Opportunities and Pitfalls

One cost-effective alternative delegates judging to online micro-task platforms (e.g., Amazon Mechanical Turk). However, empirical literature indicates that crowdsourced annotations suffer from:

- Substantially higher variance compared to domain experts;
- Inconsistent quality and exposure to adversarial spamming;
- A requirement for robust filtering mechanisms (majority voting, worker reputation scoring, Cohen’s κ inter-rater reliability thresholds).

A more recent and increasingly adopted alternative is **LLM-as-a-Judge**:

- **LLM-as-a-Judge:** State-of-the-art language models (such as GPT-4-class models) are prompted with structured evaluation rubrics to assign relevance grades to query-document pairs. This approach offers substantial *economic scalability*—millions of judgments can be obtained at a fraction of the cost of human crowdworkers. However, it introduces well-documented *intrinsic biases*: **length bias** (longer, more verbose responses tend to receive higher scores regardless of content quality), **self-preference bias** (a model tends to rate its own outputs more favorably), and **position bias** (the order in which candidates are presented influences the judgment). Active research is ongoing to quantify and mitigate these distortions.

2.4.4 The TREC Infrastructure

Organized by the National Institute of Standards and Technology (NIST), the **Text REtrieval Conference (TREC)** provides the foundational infrastructure for large-scale IR experimentation. TREC is organized into specialized thematic *tracks* reflecting distinct information access tasks (Web search, biomedical IR, conversational search, deep learning retrieval). The output produced by a participant system executing a retrieval task on a test collection is termed a **Run**.

Table 2.1 documents the dimensional evolution of classical benchmark collections across Information Retrieval history [1], emphasizing the massive scale increase that necessitated the creation of TREC.

Collection	Documents (N)	Queries	Size (MB)	Terms/Doc	Assessments (Q-D)
ADI	82	35	—	—	—
AIT	2 109	14	2	400.0	> 10 000
CACM	3 204	64	2	24.5	—
CISI	1 460	112	2	46.5	—
Cranfield	1 400	225	2	53.1	Exhaustive
LISA	5 872	35	3	—	—
Medline	1 033	30	1	—	—
NPL	11 429	93	3	—	—
OSHMED	348 566	106	400	250.0	16 140
Reuters	21 578	672	28	131.0	—
TREC (Typical)	740 000	200	2 000 (2 GB)	89–3 543	> 100 000

Table 2.1: Comparative overview of classical historical evaluation test corpora.

As evident from the empirical data, early corpora from the 1960s and 1970s (such as Cranfield and Medline) comprised only a few thousand records and 1–2 MB, allowing exhaustive manual judging. With the advent of realistic Web-scale corpora containing hundreds of thousands of documents

and gigabytes of text like TREC, the combinatorial evaluation space exploded, mandating selective sampling strategies.

2.4.5 The Pooling Technique

To select which documents human assessors should judge without evaluating the entire corpus, TREC standardizes the **Pooling** methodology (Figure 2.2):

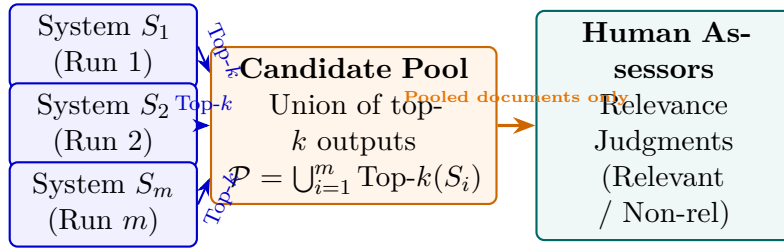


Figure 2.2: The Pooling technique for selecting candidate documents for human judgment.

1. Collect ranked result lists from m diverse retrieval systems for a given topic.
2. Extract the top- k ranked documents from each run (typically $k = 50$ or $k = 100$).
3. Take the **union** of these candidate sets, deduplicating documents to build pool $\mathcal{P}$.
4. Human judges assess relevance strictly for documents inside $\mathcal{P}$.

The Unjudged Document Assumption

All documents in the corpus not present in the pool (*unpooled*) are formally assumed to be **non-relevant** ($r = 0$). While this assumption may theoretically penalize innovative systems surfacing relevant documents overlooked by historical pool contributors, NIST experiments confirm that sufficiently diverse pools maintain stable, fair, and reliable system rankings.

2.5 Information Need vs. Query

A crucial principle in evaluation separates the underlying need from its textual expression:

- **Information Need:** The user’s actual cognitive desire. Example: “*My swimming pool bottom is becoming black and needs to be cleaned*”.
- **Query:** The keyword shorthand submitted to the system. Example: “pool cleaner”.

Property 2.5.1 (Evaluation Principle). *Document relevance must always be assessed with respect to the user’s underlying **Information Need**, never purely against the superficial keywords of the query string.*

2.6 Unranked Retrieval Measures

In unranked evaluation contexts (e.g., Boolean retrieval), the engine returns an unordered set of documents. Performance is derived from the standard contingency table (Table 2.2):

	Document Relevant	Document Non-Relevant
Document Retrieved	True Positive (TP)	False Positive (FP)
Document Not Retrieved	False Negative (FN)	True Negative (TN)

Table 2.2: Contingency matrix for binary query-document classification.

2.6.1 Precision, Recall, and the Precision-Recall Trade-off

Definition 2.6.1 (Precision (P)). The fraction of retrieved documents that are relevant:

$$P = \frac{TP}{TP + FP} = P(\text{relevant} \mid \text{retrieved}) \quad (2.1)$$

Definition 2.6.2 (Recall (R)). The fraction of relevant documents in the collection that were successfully retrieved:

$$R = \frac{TP}{TP + FN} = P(\text{retrieved} \mid \text{relevant}) \quad (2.2)$$

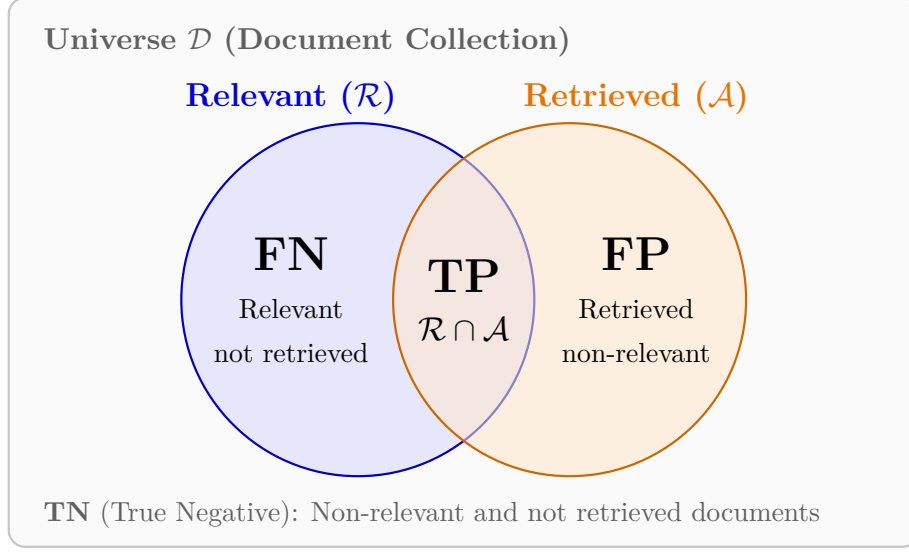


Figure 2.3: Set-theoretic Venn diagram of document retrieval and confusion regions.

An inherent **trade-off** governs Precision and Recall:

- Returning the entire document repository (Retrieved = $\mathcal{D}$) guarantees $R = 100\%$, but drops precision near zero ($P \approx 0$).
- Conversely, returning only the single most confident document can yield $P = 100\%$, at the expense of dismal recall.

2.6.2 The Harmonic Mean: F-Measure (F_1 -Score)

To balance Precision and Recall into a single scalar metric that penalizes extreme disparity, the **F-Measure** calculates their **harmonic mean**:

$$F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (2.3)$$

Property 2.6.1 (Harmonic Mean Properties). *The harmonic mean exhibits key mathematical properties for evaluation:*

1. *It is always upper-bounded by both the geometric and arithmetic means:*

$$\frac{2}{\frac{1}{a} + \frac{1}{b}} \leq \sqrt{a \cdot b} \leq \frac{a + b}{2}$$

2. When the two operands diverge significantly, the harmonic mean is heavily drawn toward the **minimum** rather than the arithmetic average.

Example 2.6.1 (Edge Behavior of F-Measure). Consider a degenerate retrieval system returning all documents: $R = 1.0$ and $P = 0.0001$.

- The arithmetic mean yields an artificially inflated score:

$$\frac{1.0 + 0.0001}{2} \approx 0.50 \quad (50\%)$$

- The harmonic mean (F_1) penalizes the system:

$$F_1 = \frac{2 \cdot 1.0 \cdot 0.0001}{1.0 + 0.0001} \approx 0.0002 \quad (0.02\%)$$

This illustrates why the harmonic mean is mathematically favored over the arithmetic mean.

2.6.3 The Weighted F_β -Measure

When asymmetric priority is given to Precision versus Recall, a weighting parameter $\alpha \in [0, 1]$ is introduced:

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} \quad (2.4)$$

Setting $\alpha = \frac{1}{1+\beta^2}$ leads to the standard F_β formulation:

$$F_\beta = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R} \quad (2.5)$$

- Values of $\beta < 1$ (e.g., $\beta = 0.5$) emphasize **Precision** over Recall (favored in Web search).
- Values of $\beta > 1$ (e.g., $\beta = 2.0$) emphasize **Recall** over Precision (crucial in legal discovery, patent search, and clinical meta-analyses).

2.7 Rank-Based Measures (Binary Relevance)

Modern search engines return ranked lists ordered by descending relevance scores. Furthermore, Web users rarely look beyond the first page of results ($k = 10$ or $k = 30$).

2.7.1 Precision@k ($P@k$) and Recall@k ($R@k$)

Definition 2.7.1 (Precision@k). Given a fixed rank cutoff $k \in \mathbb{N}^+$, Precision@k is the fraction of relevant documents within the top k positions:

$$P@k = \frac{1}{k} \sum_{i=1}^k \text{rel}_i \quad (2.6)$$

where $\text{rel}_i \in \{0, 1\}$ represents the binary relevance of the document at rank i .

Example 2.7.1 (Computing Precision@k). Consider a ranked list with binary relevance sequence for the top 5 positions $\langle 1, 1, 0, 0, 1 \rangle$:

- $P@1 = \frac{1}{1} = 1.00$
- $P@2 = \frac{2}{2} = 1.00$
- $P@3 = \frac{2}{3} \approx 0.66$
- $P@4 = \frac{2}{4} = 0.50$ (note that $P@4 < P@3$)
- $P@5 = \frac{3}{5} = 0.60$

The Distortion of $P@k$ When Averaged Across Queries

While $P@k$ is intuitive for estimating top-page result quality, it suffers from a critical methodological flaw when averaged across heterogeneous queries:

- If a specific query q_A possesses only $R_A = 2$ relevant documents in the entire corpus, the theoretical maximum possible $P@10(q_A)$ is capped at $\frac{2}{10} = 0.20$, even if a flawless engine places both relevant documents at ranks 1 and 2!
- Averaging raw $P@10$ across queries penalizes the retrieval engine for the intrinsic scarcity of relevant documents for specific information needs, rather than algorithmic ranking errors.

Similarly, the complementary measure *Recall@k* ($R@k$) evaluates the fraction of all relevant documents recovered within rank k :

$$\text{Recall@}k = \frac{1}{R} \sum_{i=1}^k \text{rel}_i \quad (2.7)$$

2.7.2 Average Precision (AP)

Average Precision (AP) summarizes the entire Precision-Recall curve into a single scalar score for a given query, evaluating instantaneous precision at each rank position where recall increases.

Definition 2.7.2 (Average Precision). For a ranking of N documents corresponding to a query with R total relevant documents in the collection:

$$AP = \frac{1}{R} \sum_{k=1}^N P@k \cdot \text{rel}_k \quad (2.8)$$

Property 2.7.1 (Geometric Interpretation and the Role of $1/R$). *Average Precision carries a direct analytical meaning:*

1. Each time a relevant document is encountered at rank k , recall increases by a discrete increment $\Delta\text{Recall} = \frac{1}{R}$. AP accumulates the rectangular areas $P@k \cdot \Delta\text{Recall}$, representing the **discrete numerical approximation of the Area Under the Precision-Recall Curve** (AUC-PR).
2. Normalizing by R (the total number of relevant documents in the corpus) ensures that relevant documents omitted from the top- N candidate pool implicitly contribute zero precision, heavily penalizing systems with incomplete recall.

	Search Engine Output Order →				
Rank Position (k):	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Document:	Rel	NonRel	Rel	NonRel	Rel
Precision@k:	1.00	—	0.67	—	0.60

Figure 2.4: Schematic representation of Average Precision calculation over 5 documents with $R = 3$ total relevant items.

Example 2.7.2 (Basic Calculation with $R = 3$ Relevant Documents). Consider the ranking scenario in Figure 2.4 with $R = 3$ total relevant documents appearing at ranks $k_1 = 1$, $k_2 = 3$, and $k_3 = 5$:

$$P@1 = \frac{1}{1} = 1.00, \quad P@3 = \frac{2}{3} \approx 0.667, \quad P@5 = \frac{3}{5} = 0.60$$

The resulting Average Precision is:

$$AP = \frac{1}{3} \left(1.00 + \frac{2}{3} + \frac{3}{5} \right) = \frac{1}{3} (2.267) \approx 0.756 \quad (75.6\%) \quad (2.9)$$

Example 2.7.3 (Comparative Analysis Between Two Rankings). Consider a query associated with $R = 6$ relevant documents in the collection. Two competing search algorithms output the following top-10 ranked results:

- **Ranking #1:** Relevant items concentrated early in the list:

$\langle \text{Rel}, \text{NonRel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{NonRel}, \text{NonRel}, \text{NonRel}, \text{Rel} \rangle$

- **Ranking #2:** Relevant items delayed and scattered:

$\langle \text{NonRel}, \text{Rel}, \text{NonRel}, \text{NonRel}, \text{Rel}, \text{Rel}, \text{Rel}, \text{NonRel}, \text{Rel}, \text{Rel} \rangle$

Table 2.3 presents the step-by-step evolution of Precision and Recall across all 10 ranks.

Rank (k)	Ranking #1			Ranking #2		
	Item	Recall	Precision	Item	Recall	Precision
1	Rel	0.17	1.00	NonRel	0.00	—
2	NonRel	0.17	—	Rel	0.17	0.50
3	Rel	0.33	0.67	NonRel	0.17	—
4	Rel	0.50	0.75	NonRel	0.17	—
5	Rel	0.67	0.80	Rel	0.33	0.40
6	Rel	0.83	0.83	Rel	0.50	0.50
7	NonRel	0.83	—	Rel	0.67	0.57
8	NonRel	0.83	—	NonRel	0.67	—
9	NonRel	0.83	—	Rel	0.83	0.56
10	Rel	1.00	0.60	Rel	1.00	0.60
Average Precision (AP)			0.78	0.52		

Table 2.3: Detailed comparative evaluation of Average Precision between Ranking #1 and Ranking #2.

The resulting AP scores evaluate to:

$$AP(\text{Ranking \#1}) = \frac{1.00 + 0.67 + 0.75 + 0.80 + 0.83 + 0.60}{6} = \frac{4.65}{6} \approx \mathbf{0.78}$$

$$AP(\text{Ranking \#2}) = \frac{0.50 + 0.40 + 0.50 + 0.57 + 0.56 + 0.60}{6} = \frac{3.13}{6} \approx \mathbf{0.52}$$

Although both systems achieve identical overall recall ($R = 100\%$ across 10 retrieved documents), Ranking #1 obtains a substantially higher score (0.78 vs 0.52) because it rewards front-loading relevant items.

2.7.3 Mean Average Precision (MAP)

To quantify overall effectiveness across a suite of $|\mathcal{Q}|$ benchmark queries, we compute the arithmetic mean of their respective AP scores:

Definition 2.7.3 (Mean Average Precision).

$$MAP = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} AP(q) \quad (2.10)$$

Property 2.7.2 (Macro-Averaging vs. Micro-Averaging). *MAP formally implements a **Macro-Averaging** scheme: each query carries identical statistical weight $1/|\mathcal{Q}|$.*

- *In contrast, Micro-Averaging (pooling all instantaneous precisions across all queries and dividing by the total sum of relevant documents) would be overwhelmingly dominated by a handful of broad queries (e.g., "movies" or "news", which often match thousands of relevant documents), drowning out performance on specific long-tail queries.*
- *Macro-Averaging ensures that every individual information need receives equal representation in the final benchmark metric.*

Example 2.7.4 (Computing MAP Over Two Queries). Consider an evaluation suite consisting of two distinct queries q_1 and q_2 :

- **Query 1 ($R_1 = 5$ total relevant documents):** The engine returns 5 relevant items at ranks $k \in \{1, 3, 6, 9, 10\}$ with precisions 1.00, 0.67, 0.50, 0.44, and 0.50:

$$AP(q_1) = \frac{1.00 + 0.67 + 0.50 + 0.44 + 0.50}{5} = \frac{3.11}{5} \approx \mathbf{0.62}$$

- **Query 2 ($R_2 = 3$ total relevant documents):** The engine returns 3 relevant items at ranks $k \in \{2, 5, 7\}$ with precisions 0.50, 0.40, and 0.43:

$$AP(q_2) = \frac{0.50 + 0.40 + 0.43}{3} = \frac{1.33}{3} \approx \mathbf{0.44}$$

The overall Mean Average Precision across both queries is:

$$MAP = \frac{AP(q_1) + AP(q_2)}{2} = \frac{0.62 + 0.44}{2} = \mathbf{0.53} \quad (2.11)$$

2.7.4 Truncated Ranking Measures: AP@k and MAP@k

In Web search and Recommender Systems, evaluating AP across an entire collection is impractical (the true total count R of relevant items is unknown) and disconnected from user browsing habits. Engines thus adopt rank-truncated formulations (**MAP@K**):

Definition 2.7.4 (AP@K and MAP@K). Given a fixed rank cutoff K (typically $K = 10$ or $K = 20$), Average Precision at K for query q_i is defined as:

$$AP@K(q_i) = \frac{1}{K_i} \sum_{k=1}^K P@k(q_i) \cdot \text{rel}(q_i, k) \quad (2.12)$$

where the normalizing denominator K_i reflects the number of relevant items within the top K :

$$K_i = \sum_{k=1}^K \text{rel}(q_i, k) \quad (\text{or } K_i = \min(R_i, K)) \quad (2.13)$$

The global **MAP@K** across query suite $\mathcal{Q}$ evaluates to:

$$MAP@K(\mathcal{Q}) = \frac{1}{|\mathcal{Q}|} \sum_{q_i \in \mathcal{Q}} AP@K(q_i) \quad (2.14)$$

This formulation resolves the fundamental flaw of raw $P@K$: by normalizing by K_i instead of a rigid K , a query with only 2 relevant documents that places both in the top 2 slots attains $AP@10 = 1.0$ rather than being artificially capped at 0.20.

2.7.5 Mean Reciprocal Rank (MRR)

In tasks such as known-item search, navigational queries (e.g., locating an institutional homepage), or factoid retrieval, the user is interested solely in finding the **first correct document**. User effort is inversely proportional to the rank of that first hit:

Definition 2.7.5 (Reciprocal Rank and MRR). Let $\text{rank}_1(q)$ denote the rank position of the first relevant document returned for query q . The **Reciprocal Rank** (RR) is:

$$RR(q) = \frac{1}{\text{rank}_1(q)} \quad (2.15)$$

The **Mean Reciprocal Rank** (MRR) across all queries is:

$$MRR = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{\text{rank}_1(q)} \quad (2.16)$$

If the first relevant hit appears at rank 1, the score is 1.0; if at rank 2, it drops to 0.5; if delayed to rank 10, it falls to 0.1.

2.8 Beyond Binary Relevance: Graded Relevance and NDCG

In practical Web and commercial applications, binary judgments prove inadequate. Consider the following real-world Google search query:

Query: "information retrieval unipi"

A commercial SERP returns pages exhibiting distinct qualitative levels of utility:

1. **Course DidaWiki (Instructor, CFU, syllabus, lecture notes):** The primary resource fully satisfying user search intent $\implies$ **Highly Relevant** ($r = 3$).
2. **Official UniPi Valutami Portal (Exam registration and administrative syllabus):** Authoritative institutional record but lacks study material $\implies$ **Relevant** ($r = 2$).
3. **Master's Degree PDF Brochure:** Contains only a brief mention of the course $\implies$ **Marginally Relevant** ($r = 1$).
4. **Unrelated articles or campus news:** $\implies$ **Irrelevant** ($r = 0$).

While all top three pages are pertinent, user satisfaction depends heavily on displaying the most informative page at rank 1. To capture this nuance, IR formalizes multi-grade relevance scales ($r \in \{0, 1, 2, 3\}$ or $\{0, \dots, 4\}$).

The **Discounted Cumulative Gain** (DCG) framework, introduced by Järvelin and Kekäläinen [8], rests on two assumptions:

1. Highly relevant documents are much more useful than marginally relevant ones.
2. The lower a document appears in the ranking, the less likely the user is to examine it; hence, gain must be discounted with descending rank.

2.8.1 Cumulative Gain (CG) and Discounted Cumulative Gain (DCG)

Definition 2.8.1 (Cumulative Gain at Rank p). Given a vector of graded relevance values $\mathbf{r} = [r_1, r_2, \dots, r_p]$, the Cumulative Gain is the unweighted sum:

$$CG_p = \sum_{i=1}^p r_i \quad (2.17)$$

CG_p is position-insensitive: swapping a highly relevant document at rank 1 with one at rank 10 leaves CG_p unchanged.

Definition 2.8.2 (Discounted Cumulative Gain (Standard Formulation)). Standard DCG applies a logarithmic discount based on rank i :

$$DCG_p = r_1 + \sum_{i=2}^p \frac{r_i}{\log_2(i)} \quad (2.18)$$

Behavioral Meaning of the Logarithmic Discount

The discount factor $\frac{1}{\log_2(i)}$ reflects the exponential decay of user visual attention under cascade reading models:

- At rank $i = 4$, the discount factor is $\frac{1}{\log_2(4)} = \frac{1}{2}$: perceived informative utility is cut in half relative to rank 1.
- At rank $i = 8$, the factor drops to $\frac{1}{\log_2(8)} = \frac{1}{3}$.
- At rank $i = 16$, the factor falls to $\frac{1}{\log_2(16)} = \frac{1}{4}$.

Commercial search engines (Google, Bing) frequently utilize the **exponential formulation**:

$$DCG_p^{\text{exp}} = \sum_{i=1}^p \frac{2^{r_i} - 1}{\log_2(i + 1)} \quad (2.19)$$

Summation Index Clarification

Note the index shift in the logarithm denominator:

- In Equation (2.18), r_1 is separated to prevent division by $\log_2(1) = 0$, while the summation starts at $i = 2$ with $\log_2(i)$.
- In Equation (2.19), the summation starts directly at $i = 1$ by taking $\log_2(i + 1)$ ($\log_2(1 + 1) = \log_2(2) = 1$).
- Furthermore, numerator $2^{r_i} - 1$ non-linearly rewards top-tier relevance ($r = 3 \implies 2^3 - 1 = 7$, vs. $r = 1 \implies 2^1 - 1 = 1$), severely penalizing any demotion of stellar results.

2.8.2 Normalized Discounted Cumulative Gain

Because different queries have varying numbers of relevant documents, raw DCG_p scores cannot be compared across queries. We normalize by the **Ideal DCG** ($IDCG_p$), obtained by computing the DCG of the ideal ranking sorted by relevance descending:

$$NDCG_p = \frac{DCG_p}{IDCG_p} \in [0, 1] \quad (2.20)$$

2.8.3 Complete Numerical Walkthrough

Consider the following 10-document ranking, judged on a scale of 0–3:

$$\mathbf{r} = [3, 2, 3, 0, 0, 1, 2, 2, 3, 0]$$

The ideal sorted ranking is:

$$\mathbf{r}^* = [3, 3, 3, 2, 2, 2, 1, 0, 0, 0]$$

Table 2.4 summarizes the step-by-step progression across all 10 ranks.

2.8.4 Comparing Alternative Ranking Functions

Consider 4 documents $\{d_1, d_2, d_3, d_4\}$ with graded assessments:

$$r(d_1) = 0, \quad r(d_2) = 1, \quad r(d_3) = 2, \quad r(d_4) = 2$$

i	r_i	Discount	Discounted Gain	$DCG@i$	r_i^*	Ideal Gain	$IDCG@i$	$NDCG@i$
1	3	1.000	3.00	3.00	3	3.00	3.00	1.00
2	2	$\log_2(2) = 1.000$	2.00	5.00	3	3.00	6.00	0.83
3	3	$\log_2(3) = 1.585$	1.89	6.89	3	1.89	7.89	0.87
4	0	$\log_2(4) = 2.000$	0.00	6.89	2	1.00	8.89	0.76
5	0	$\log_2(5) = 2.322$	0.00	6.89	2	0.86	9.75	0.71
6	1	$\log_2(6) = 2.585$	0.39	7.28	2	0.77	10.52	0.69
7	2	$\log_2(7) = 2.807$	0.71	7.99	1	0.36	10.88	0.73
8	2	$\log_2(8) = 3.000$	0.67	8.66	0	0.00	10.88	0.80
9	3	$\log_2(9) = 3.170$	0.95	9.61	0	0.00	10.88	0.88
10	0	$\log_2(10) = 3.322$	0.00	9.61	0	0.00	10.88	0.88

Table 2.4: Detailed trace of $DCG@i$, $IDCG@i$, and $NDCG@i$ computation for ranks 1 through 10.

- **Ground Truth (GT):** Ordering $\langle d_4, d_3, d_2, d_1 \rangle$ with relevance vector $[2, 2, 1, 0]$:

$$IDCG_4 = 2 + \frac{2}{\log_2(2)} + \frac{1}{\log_2(3)} + 0 = 2 + 2 + 0.6309 = 4.6309$$

$$NDCG_{GT} = 1.00$$

- **Ranking Function 1 (RF_1):** Ordering $\langle d_3, d_4, d_2, d_1 \rangle$ with relevance $[2, 2, 1, 0]$. Because the first two documents share identical relevance ($r = 2$), their permutation incurs no positional penalty:

$$DCG_{RF_1} = 2 + \frac{2}{\log_2(2)} + \frac{1}{\log_2(3)} = 4.6309$$

$$NDCG_{RF_1} = \frac{4.6309}{4.6309} = 1.00$$

- **Ranking Function 2 (RF_2):** Ordering $\langle d_3, d_2, d_4, d_1 \rangle$ with relevance $[2, 1, 2, 0]$. Here, d_2 ($r = 1$) erroneously outranks d_4 ($r = 2$):

$$DCG_{RF_2} = 2 + \frac{1}{\log_2(2)} + \frac{2}{\log_2(3)} + 0 = 2 + 1 + 1.2619 = 4.2619$$

$$NDCG_{RF_2} = \frac{DCG_{RF_2}}{IDCG_4} = \frac{4.2619}{4.6309} \approx 0.9203$$

This illustrates how NDCG mathematically penalizes rank inversions in direct proportion to the relevance difference.

Chapter 3

Algorithmic Efficiency on Modern CPU Architectures

3.1 The Role of Efficiency in Information Retrieval

The development of information processing systems unfolds along two complementary dimensions that are often in tension with one another:

1. **Effectiveness:** maximizes end-user satisfaction, focusing on the semantic quality of results, ranking, and natural language processing, typically using high-level abstraction tools (Python, Machine Learning frameworks).
2. **Efficiency:** minimizes infrastructural operating costs, designing algorithms optimized down to the last CPU clock cycle and the last bit of occupied memory.

No single professional figure is required to master both extremes; however, those who develop at a high level must understand the physical constraints of the underlying hardware to avoid designing impractical solutions, while those who optimize at a low level must preserve the structure and semantics of the data so as not to compromise its effectiveness.

3.1.1 The Three Pillars of Efficiency in IR Systems

User experience. In web-scale systems, a query response must be delivered within an infinitesimal fraction of a second. Users perceive any delay beyond a few tenths of a second: if results are slow or irrelevant, they abandon

the service in favor of a competitor. Algorithmic efficiency does not only serve to respond faster, but allows more sophisticated and effective ranking and filtering algorithms to run *within the same predetermined time budget* (typically a hard ceiling of 100–200 ms).

Operational scalability. Modern systems handle billions of documents, multimedia items, queries, and logs generated every second. The number of machines required is approximately:

$$\text{Number of machines} = \left\lceil \frac{\text{Total data volume}}{\text{Single-node operating capacity}} \right\rceil.$$

If the data volume doubles, a well-designed system requires at most twice as many machines (linear scalability); an inefficient algorithm can instead require superlinear cost growth (e.g., quadrupling servers), which is unsustainable at scale. To meet required latencies, active data must reside in RAM and avoid disk access: this requires algorithms that are efficient not only in time but also in space, via compression techniques. Furthermore, since data is replicated across multiple machines for fault tolerance, every memory inefficiency is amplified by the infrastructure’s replication factor.

Environmental and economic sustainability. Reducing the number of physical machines directly cuts data center energy consumption. High-density servers dissipate large amounts of heat, and much of the energy is consumed by cooling systems; for this reason many large-scale data centers are located in cold regions or near bodies of water (*free cooling*). A marginal computational saving, even just 3%, generates significant economic savings and reduces the carbon footprint.

3.2 The Memory Hierarchy and Locality Principles

3.2.1 The Latency Gap Between CPU and Memory

In the classical theoretical computational model (abstract RAM machine), access to any memory location has constant unit cost, $\mathcal{O}(1)$. On real physical machines, the CPU’s arithmetic units operate at speeds enormously higher than main memory (DRAM), and there exists a hierarchy of intermediate levels (*caches*) that approximates fast access for frequently used data.

Hardware level	Typical latency	CPU cycles	Typical capacity
CPU instruction (ALU)	0.5–5 ns	1–2	internal registers
L1I / L1D Cache	0.5 ns	≈ 1	32 KB per core
L2 Cache	1.5 ns	≈ 3 –4	256 KB (up to 2 MB)
L3 Cache (LLC)	5.0 ns	≈ 10 –15	16 MB (up to 64 MB+)
Main RAM (DRAM)	80.0 ns	≈ 160 –200+	64 GB+
Disk / Storage	500 000 ns (0.5 ms)	millions	8 TB+

Table 3.1: Quantitative comparison of latency and capacity across memory hierarchy levels.

Definition 3.2.1 (*Memory-Bound Algorithm*). An algorithm that accesses main RAM in a disordered fashion pays ≈ 80 ns per access, an interval during which the CPU could execute hundreds of instructions. Such an algorithm is called **memory-bound**: execution time is dominated by waiting for data transfer from DRAM, and the CPU spends most of its time stalled, making the optimization of arithmetic operation count irrelevant.

3.2.2 Physical On-Chip Architecture

The L1 cache is integrated close to the decode unit and execution units; to avoid bus conflicts it is physically split into **L1I** (*Instruction Cache*, holding fetched machine code for execution) and **L1D** (*Data Cache*, holding operands for `load/store` instructions). The L2 cache is integrated within the single core and acts as a low-latency buffer; the L3 cache is external to individual cores but internal to the package, and is shared among all cores of the machine.

The Cache Line as the Minimum Transfer Unit

RAM and the cache hierarchy never transfer single bytes or isolated individual variables: the minimum hardware transfer unit is the **cache line**, typically $B = 64$ bytes. If a program needs a 64-bit integer (8 bytes) located at address A , the hardware is forced to load the entire aligned block

$$[\lfloor A/64 \rfloor \times 64, \lfloor A/64 \rfloor \times 64 + 63].$$

If the program reads that single value and then jumps to a random address, the remaining 56 bytes (87.5% of the memory traffic) are wasted.

3.2.3 Temporal and Spatial Locality

A program effectively exploits the memory hierarchy only if it respects at least one of the two forms of locality:

- **Temporal locality:** when a datum accessed at time t is likely to be accessed again shortly thereafter (e.g., loop counters, frequently reused variables).
- **Spatial locality:** when accessing an address makes it likely that nearby addresses will be accessed soon after (e.g., sequential array traversal), which hardware exploits through the cache line and the **prefetcher** (a unit that detects linear access patterns and speculatively loads upcoming cache lines before they are explicitly requested).

Definition 3.2.2 (Sequential Access vs. *Pointer Chasing*). A **sequential** access pattern visits contiguous memory addresses in increasing order, without isolated loops or disjoint sub-cycles. **Pointer chasing** instead follows a chain of pointers whose target addresses are unpredictable in advance and typically scattered across memory, defeating spatial locality and prefetching.

Both functions below compute the sum of all elements of the vector, $\sum_{i=0}^{N-1} v[i] = \frac{(N-1)N}{2}$.

Listing 3.1: Linear sequential scan.

```
1 pub fn sum(v: &[usize]) -> usize {
2     let mut sum = 0;
3     for &value in v.iter() {
4         sum += value;
5     }
6     sum
7 }
```

Listing 3.2: Scan via random jumps (*pointer chasing*).

```
1 pub fn sum_by_jumping(v: &[usize]) -> usize {
2     let mut sum = 0;
3     let mut k = 0;
4     for _ in 0..v.len() {
5         sum += v[k];
6         k = v[k]; // the value read becomes the index
7                   for the next iteration
8     }
9     sum
10 }
```

In the theoretical RAM model both functions execute N iterations with one addition per step: they have identical asymptotic complexity $\mathcal{O}(N)$. On real hardware, measuring the average time per element as array size varies, marked divergences emerge instead.

N (elements)	Size	Level	<code>sum()</code>	<code>sum_by_jumping()</code>	Slowdown
256	2 KiB	L1 (32 KB)	0.1 ns	1.3 ns	13×
8 192	64 KiB	L2 (256 KB)	0.1 ns	2.5 ns	25×
262 144	2 MiB	L3 (16 MB)	0.2 ns	13.4 ns	67×
16 777 216	128 MiB	RAM (DRAM)	0.4 ns	84.9 ns	212×

Table 3.2: Average time per element for `sum()` and `sum_by_jumping()` as array size varies.

3.2.4 Analysis of Physical Causes

Why does `sum()` only vary between 0.1 and 0.4 ns? With 64-bit `usize` integers (8 bytes), a 64-byte cache line holds $64/8 = 8$ contiguous elements: there is full spatial locality. Accessing a new line incurs an initial miss, but the following 7 accesses occur at near-zero cost from cache, and the stream prefetcher loads subsequent lines ahead of execution. Denoting by T_{comp} the computation time and by T_{load} the amortized transfer time, the observed cost per element follows

$$\text{Observed latency} \approx \max(T_{\text{comp}}, T_{\text{amortized_load}}),$$

almost entirely masked by computation; the increase from 0.1 to 0.4 ns at 128 MiB is due solely to saturation of the bus bandwidth toward RAM.

Why does `sum_by_jumping()` reach 84.9 ns? Since the permutation is random, the address $k_{t+1} = v[k_t]$ points to an arbitrary location: the probability that it falls in the same cache line is negligible, so for every 8-byte element read the remaining 56 transferred bytes are discarded (total absence of spatial locality). With no linear or constant stride, prefetching is inhibited. Moreover, the CPU must know the index of the next read before it can issue it: this is a **serial dependency** that prevents any overlap. When the array size exceeds L3 capacity ($128 \text{ MiB} \gg 16 \text{ MB}$), every access becomes a global cache miss to DRAM:

Time per element $\approx$ physical DRAM access latency $\approx 80\text{--}85 \text{ ns}$,

a value that matches exactly the 84.9 ns measured.

Example 3.2.1 (Amortizing the Cost of *Pointer Chasing* by Exploiting the Cache Line). If the jump occurs between 64-byte blocks (8 64-bit integers), but all elements of the block are summed before the next jump, the cost of a cache miss to RAM (≈ 85 ns) is paid only once every 8 addition operations:

```

1 pub fn sum_by_jumping_cache_lines(v: &[usize]) -> usize
2 {
3     let mut sum = 0;
4     let mut k = 0;
5     let num_blocks = v.len() / 8;
6     for _ in 0..num_blocks {
7         let base = k * 8;
8         // fully exploits the cache line already loaded
9         // into memory
10        sum += v[base] + v[base + 1] + v[base + 2] + v[base + 3]
11            + v[base + 4] + v[base + 5] + v[base + 6];
12        k = v[base + 7] % num_blocks; // pointer to the
13        next jump
14    }
15    sum
16 }

```

The latency is amortized over 8 elements, bringing the expected average time per value to

$$\text{Average time per value} \approx \frac{85 \text{ ns}}{8} \approx 10.6 \text{ ns},$$

an intermediate result between linear scanning and element-by-element random jumping.

Remark 3.2.1 (Eliminating the Residual Slowdown of `sum()`). The slight slowdown from 0.1 to 0.4 ns observed in `sum()` occurs because $T_{\text{comp}} \approx 0.1$ ns is lower than the time needed to feed the data stream from RAM for very large vectors (memory-bound regime). By increasing the computational complexity of the loop body (e.g. multiplications or non-trivial functions on each element) one obtains $T_{\text{comp}} \gg T_{\text{load_RAM}}$: the relation $\max(T_{\text{comp}}, T_{\text{load}})$ becomes constantly equal to T_{comp} , the regime becomes *CPU-bound*, and the time curve becomes perfectly flat as N varies.

3.3 Forms of Parallelism and Pipeline Architecture

Modern CPUs exploit multiple forms of parallelism: **multi-threading/multi-core** (multiple independent physical cores on the same processor), **pipelining** (splitting instruction execution into stages operating simultaneously on adjacent instructions), and **superscalar** execution (multiple compute units within the same core to process several independent instructions per cycle).

Embarrassingly Parallel Workloads in Search Engines

Search engines do not need to synchronize multiple cores to answer a single query: they adopt **trivial parallelism** (*embarrassingly parallel*), in which, given thousands of simultaneous users, each thread on a single core serves a specific query in complete isolation. With no shared state, lock contention and cache coherence issues are eliminated.

3.3.1 Anatomy of the Five-Stage Pipeline

Pipelining decomposes instruction execution into independent sequential stages, which in a standard architecture are five:

1. **IF** (*Instruction Fetch*): fetching the machine-code instruction from memory (L1I) via the *program counter*.
2. **ID** (*Instruction Decode*): decoding the instruction into micro-operations (μ -ops) and reading source registers.
3. **EX** (*Execute*): performing the arithmetic-logic computation in the ALUs or computing the effective memory address.
4. **MEM**: access to the memory subsystem (L1D) for load or store instructions (LOAD/STORE).
5. **WB** (*Write-Back*): writing the result to architectural registers and retiring the instruction.

3.3.2 Latency, Throughput, and Pipeline Depth

Let k be the number of pipeline stages and τ the clock cycle period. The **latency** L is the total time required for a single instruction to traverse the entire pipeline, $L = k \cdot \tau$: in modern CPUs the latency for a simple operation

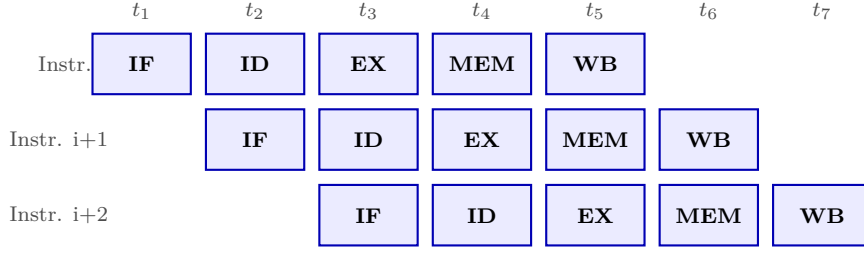


Figure 3.1: Overlap of the five pipeline stages across consecutive instructions.

typically ranges between 10 and 20 clock cycles. The **throughput** W is the number of instructions completed per unit time, and at steady state $W = 1/\tau$ instructions per second, since one instruction exits the pipeline every cycle. The period τ is dictated by the delay of the slowest stage in the circuit chain,

$$\tau \geq \max_{1 \leq i \leq k} (t_{\text{stage}_i}).$$

To raise the clock frequency (reduce τ), designers increase pipeline depth k (up to 15–20 or more stages in modern CPUs), implementing each stage with simpler, faster circuitry; conversely, increasing k inflates the penalty in case of a forced pipeline flush.

3.4 Superscalar Execution, Interleaving, and Structural Hazards

A processor is called **superscalar** when it has the architectural ability to issue and complete more than one instruction per clock cycle ($IPC > 1$, *Instructions Per Cycle*) within the same core, integrating multiple arithmetic-logic units (integer ALUs, multipliers, shift units, load/store units) served by parallel issue ports (*execution ports*). Real architectures such as Intel Haswell feature, for instance, a 56-entry decode queue, a 192-entry reorder buffer, a unified 60-entry reservation station, and 8 parallel execution ports, each specialized for classes of operations (integer ALUs, load/store, vector units, branch, etc.).

3.4.1 The Arithmetic *Interleaving* Experiment

Four Rust functions operating on a vector of 64-bit integers `&[u64]` are benchmarked, accumulating an increasing number of independent operations per iteration.

Listing 3.3: From a single reduction to four independent reductions per iteration.

```
1 pub fn do_stuff(v: &[u64]) -> u64 {
2     let mut prod = 1;
3     for &value in v.iter() {
4         prod *= value;
5     }
6     prod
7 }
8
9 pub fn do_more_stuff(v: &[u64]) -> (u64, u64) {
10     let mut prod = 1;
11     let mut sum = 0;
12     for &value in v.iter() {
13         prod *= value;
14         sum += value;
15     }
16     (prod, sum)
17 }
18
19 pub fn do_more_stuff_2(v: &[u64]) -> (u64, u64, u64) {
20     let mut prod = 1;
21     let mut sum = 0;
22     let mut xor = 0;
23     for &value in v.iter() {
24         prod *= value;
25         sum += value;
26         xor ^= value;
27     }
28     (prod, sum, xor)
29 }
30
31 #[inline]
32 pub fn do_more_stuff_3(v: &[u64]) -> (u64, u64, u64, u64)
33     {
34     let mut prod = 1;
35     let mut sum = 0;
36     let mut xor = 0;
37     let mut or = 0;
38     for &value in v.iter() {
39         prod *= value;
40         sum += value;
41         xor ^= value;
```

```

41     or |= value;
42 }
43 (prod, sum, xor, or)
44 }

```

Function	Time per iteration
do_stuff() (1 operation)	0.4 ns
do_more_stuff() (2 operations)	0.4 ns
do_more_stuff_2() (3 operations)	0.4 ns
do_more_stuff_3() (4 operations)	0.5 ns

Table 3.3: Time per iteration as the number of accumulated independent reductions grows.

3.4.2 Instruction Overlap and Port Saturation

From one to three operations the time remains unchanged at 0.4 ns because the accumulator variables have no mutual dependencies: $\text{prod}_{t+1} = \text{prod}_t \cdot v[i]$, $\text{sum}_{t+1} = \text{sum}_t + v[i]$, $\text{xor}_{t+1} = \text{xor}_t \oplus v[i]$. The superscalar core simultaneously dispatches the multiplication, addition, and XOR to three distinct execution ports: the three operations complete in the same clock cycle, and the extra work is performed at zero cost.

With the fourth operation (**or**) the time rises to 0.5 ns rather than doubling to 0.8 ns: this is a **structural hazard**, since the physical units available for parallel arithmetic-logic computation in a single cycle are saturated (only three instructions of that type can be issued simultaneously), and one of the four operations must wait. However the *out-of-order* CPU does not process iterations as isolated compartments: it implicitly unrolls the instruction window, grouping them into blocks of three, absorbing the excess operation into subsequent cycles and saturating the residual capacity of the ports. The real overhead thus amounts to a marginal fraction of a cycle ($\Delta t = +0.1$ ns) rather than a full additional cycle.

3.5 Data Hazards and Serial Dependencies

Definition 3.5.1 (Data Hazard). A **data hazard** occurs when an instruction needs an operand that is not yet available because it is still being computed by a preceding instruction. It corresponds to a **RAW** (*Read-After-Write*)

data dependency:

$$\text{Inst}_A : R_1 \leftarrow R_2 + R_3 \quad \Longrightarrow \quad \text{Inst}_B : R_4 \leftarrow R_1 \times R_5,$$

where Inst_B must stall in the pipeline until Inst_A completes its computation.

Consider code that performs the same number of operations per iteration as the previous `do_more_stuff_3`, but chains the data in a tight dependency:

Listing 3.4: Serial RAW dependency chain across four operations.

```

1 pub fn do_more_stuff_4(v: &[u64]) -> (u64, u64, u64) {
2     let mut prod = 1;
3     let mut sum = 0;
4     let mut xor = 0;
5     for &value in v.iter() {
6         sum += value; // 1. compute sum
7         prod *= sum; // 2. tightly depends on sum
8         xor ^= prod; // 3. tightly depends on prod
9         sum += xor; // 4. depends on xor and
10                // overwrites sum for the next iteration!
11     }
12     (prod, sum, xor)
13 }
```

Because of the RAW chain, the execution units cannot start any computation in parallel:

$$\text{Iteration latency} = t_{\text{add}} + t_{\text{mul}} + t_{\text{xor}} + t_{\text{add_feedback}}.$$

Since the last operation updates `sum` and the first instruction of the next iteration requires the new value of `sum`, the *out-of-order* CPU cannot start iteration $t + 1$ early: the superscalar units remain unused. Experimentally, against the 0.5 ns of `do_more_stuff_3` (independent operations), `do_more_stuff_4` measures 1.7 ns per iteration ($\approx 3.5\times$ slower), despite performing the same number of arithmetic operations.

3.5.1 Algorithmic Solution: *Interleaving* Independent Vectors

When the same function with a tight dependency must be run on two distinct vectors `V` and `West`, the traditional sequential approach (`do_more_stuff_4(V)` followed by `do_more_stuff_4(West)`) takes an overall time $2T$. If instead a single function processes both vectors within the same iteration, alternating the two independent chains:

Listing 3.5: Fusion (interleaving) of two mutually independent RAW chains.

```
1 for i in 0..n {
2     // chain for V (occupies the ports at cycle k)
3     sum_v += v[i];
4     prod_v *= sum_v;
5     xor_v ^= prod_v;
6     sum_v += xor_v;
7     // chain for West, completely independent of V!
8     sum_w += west[i];
9     prod_w *= sum_w;
10    xor_w ^= prod_w;
11    sum_w += xor_w;
12 }
```

the measured execution time is identical to that of the single vector **V** alone ($\approx T$, rather than $2T$). Operations on **West** do not depend on those on **V**: while **V**'s chain is waiting for ALU results, the free compute units process **West**'s chain, filling the dead time imposed by data dependencies at zero extra cost.

3.6 Control Hazards, Speculative Execution, and Branch Prediction

Definition 3.6.1 (Control Hazard). A **control hazard** occurs when the CPU cannot determine which instruction to fetch next (*instruction fetch*) due to the presence of a conditional branch (**if**, **branch**).

The CPU cannot stop at every condition: it therefore adopts **speculative execution**, guessing via a *branch predictor* which branch will be taken and speculatively fetching subsequent instructions to feed the pipeline. If the outcome of the condition turns out to disagree with the prediction, all speculative work completed in the intermediate stages must be discarded, with a full pipeline flush and reload from the correct branch:

Misprediction penalty = 15–20 clock cycles.

3.6.1 The Random Condition Paradox

A conditional filter is introduced into code with a dependency chain, operating on an array of random integers uniformly distributed in $[0, 99]$:

Listing 3.6: Conditional branch on random values: the CPU cannot predict the outcome.

```
1 pub fn do_more_stuff_5(v: &[u64]) -> (u64, u64, u64) {
2     let mut prod = 1;
3     let mut sum = 0;
4     let mut xor = 0;
5     for &value in v.iter() {
6         if value > 50 { // true with p = 0.5 on random
7             // values in [0, 99]
8             sum += value;
9             prod *= sum;
10            xor ^= prod;
11            sum += xor;
12        }
13    }
14    (prod, sum, xor)
}
```

Since the condition is true on average 50% of the time, one would intuitively expect a time roughly halved compared to `do_more_stuff_4` ($1.7/2 \approx 0.85$ ns). The measured time instead rises to 3.5 ns, more than twice as slow as the version without the condition.

The Real Cost Is Not Computation but Misprediction

With random values the branch predictor's accuracy is equivalent to a coin flip, $P(\text{hit}) = P(\text{miss}) = 0.50$. The average cost per iteration is then

Average cost $\approx 0.5 \times T_{\text{body}} + P(\text{miss}) \times \text{Flush penalty} \approx 0.5 \times 0.85 \text{ ns} + 0.5 \times 5.0 \text{ ns} \approx 2.9\text{--}3.5 \text{ ns}$,

having assumed a penalty of 15–20 cycles ($\approx 4\text{--}5$ ns on a 4 GHz CPU). The cost of the algorithm is no longer dictated by arithmetic operations, but is dominated by the time lost flushing and refilling the pipeline.

3.6.2 Experimental Verification: Data Ordering and Periodic Patterns

To demonstrate that the slowdown derives exclusively from branch predictor behavior, the same identical code is run while only changing the arrangement of data in the array.

With **sorted data** (all elements ≤ 50 at the start of the array and all

Configuration	Time per iteration	Predictor behavior
<code>do_more_stuff_4()</code> (no if)	1.7 ns	—
<code>do_more_stuff_5()</code> , random data	3.5 ns	unpredictable branch, miss rate $\approx 50\%$
<code>do_more_stuff_5()</code> , sorted data	1.2 ns	near-perfect prediction, single miss
<code>do_more_stuff_5()</code> , alternating data	1.1 ns	periodic sequence, miss rate $\approx 0\%$

Table 3.4: Time per iteration of `do_more_stuff_5()` as data arrangement varies.

elements > 50 at the end) the predictor assumes the branch is not taken for the whole first half and taken for the second: a single misprediction occurs at the transition point, and once flush costs are removed the time drops to 1.2 ns, lower than the 1.7 ns of the version without the condition since half the iterations do not execute the `if` body. With **alternating data** following a periodic pattern (*false, true, false, true, ...*), predictors integrate historical pattern tables (*branch history shift register*): once the periodicity is identified, the predictor anticipates the outcome at every step without errors, eliminating stalls.

3.6.3 Branchless Techniques and the Implicit Jump of for Loops

If data cannot be sorted or predicted, the only solution is to physically remove the conditional branch (*branchless code*), replacing it with algebraic expressions or conditional assembly instructions (CMOV). For example, instead of

```

1 if value > 50 {
2     sum += value;
3 }
```

the logical guard is mapped onto a scalar binary value (0 or 1), always performing the arithmetic operation:

```

1 let condition = (value > 50) as u64; // 1 if true, 0 if
   false
2 sum += value * condition;           // adds 'value' if
   true, adds '0' if false
```

The program always performs the mathematical computation, but the instruction flow is linear: there are no branches, and mispredictions are eliminated.

Remark 3.6.1 (Why the for Loop Branch Is Not a Problem). Every `for` loop ends at the machine level with a conditional jump back to the start of

the block. This branch does not degrade performance because it is highly asymmetric: for N iterations, the loop-continuation condition is true $N - 1$ consecutive times and false only upon exit. The branch predictor immediately learns the pattern (“branch taken”) and commits a single misprediction at the end of the loop, with a completely negligible impact on overall time.

3.7 Synthesis: From Hardware Constraints to Algorithmic Choices

The phenomena analyzed in this chapter – memory hierarchy and locality, pipelining, superscalar execution, *data hazards*, and *control hazards* – are not negligible implementation details, but physical constraints that determine the actual performance behavior of large-scale Information Retrieval algorithms. In particular:

- data structures (posting lists, inverted indexes) should be designed to favor sequential, contiguous access, in order to exploit spatial locality and hardware prefetching;
- when possible, it is beneficial to express multiple independent reductions within the same loop to saturate superscalar execution ports at marginal cost;
- tight (RAW) dependency chains between accumulator variables should be identified and, where possible, decoupled or interleaved with independent computations from other data streams;
- unpredictable conditional branches on data-dependent conditions should be replaced with branchless techniques whenever the misprediction cost outweighs the extra computation.

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